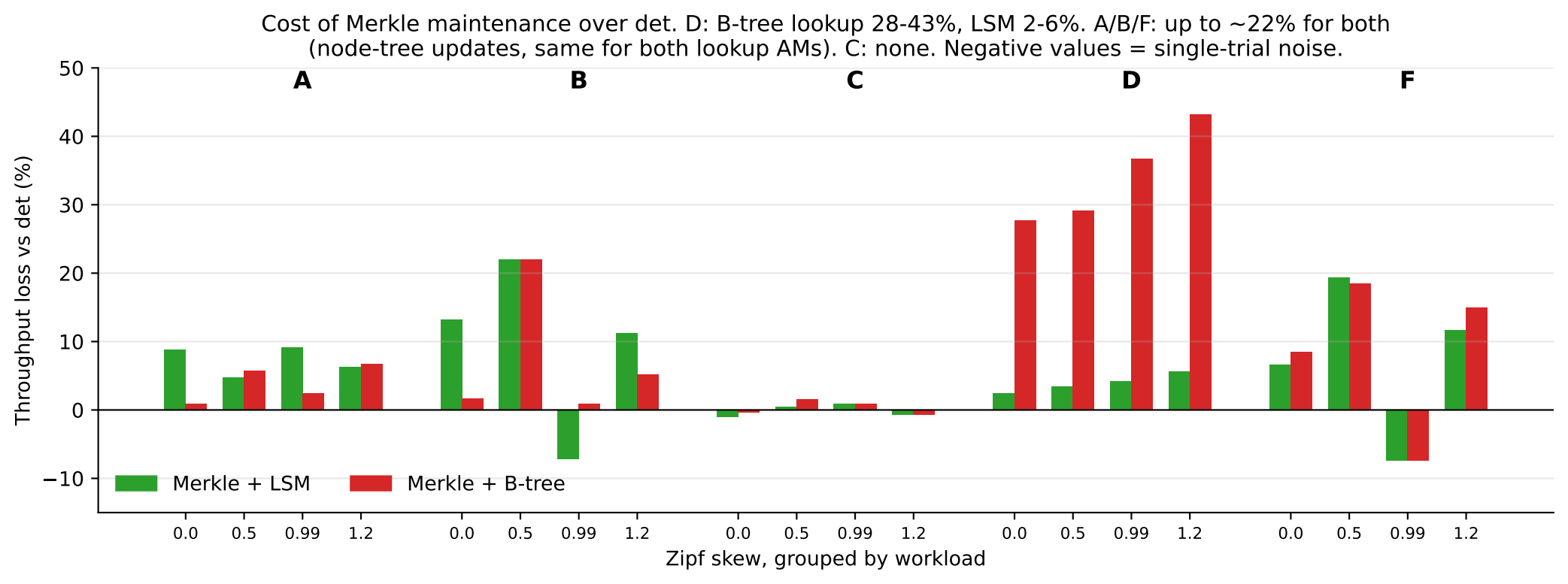
Cost of Merkle maintenance over det. D: B-tree lookup 28-43%, LSM 2-6%. A/B/F: up to ~22% for both
(node-tree updates, same for both lookup AMs). C: none. Negative values = single-trial noise.
A
B
C
D
F
Throughput loss vs det (%)
50
40
30
20
10
0
−10
Merkle + LSM
Merkle + B-tree
0.0
0.5
0.99
1.2
Zipf skew, grouped by workload